

Synergistic Buried Interface Regulation of Tin–Lead Perovskite Solar Cells via Co-Self-Assembled Monolayers

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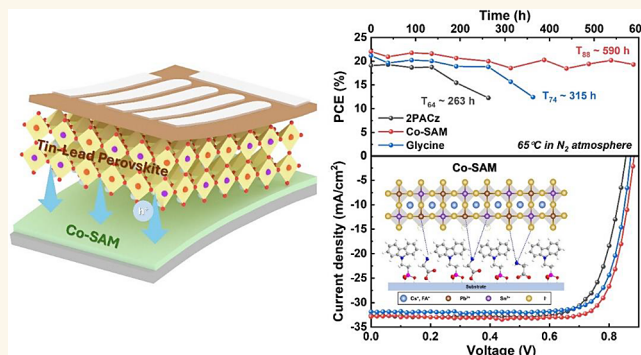
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ABSTRACT: Tin–lead (Sn–Pb) perovskite solar cells (PSCs) hold considerable potential for achieving efficiencies near the Shockley–Queisser (S–Q) limit. Notably, the inverted structure stands as the preferred fabrication method for the most efficient Sn–Pb PSCs. In this regard, it is imperative to implement a strategic customization of the hole selective layer to facilitate carrier extraction and refine the quality of perovskite films, which requires effective hole selectivity and favorable interactions with Sn–Pb perovskites. Herein, we propose the development of Co-Self-Assembled Monolayers (Co-SAM) by integrating both [2-(9H-carbazol-9-yl)ethyl]-phosphonic acid (2PACz) and glycine at the buried contacts. The one-step deposition process employed in the fabrication of the Co-SAM ensures uniform coverage, resulting in a homogeneous surface potential. This is attributed to the molecular interactions occurring between 2PACz and glycine in the processing solution. Furthermore, the amine ($-\text{NH}_2$) and ammonium ($-\text{NH}_3^+$) groups in glycine effectively passivate Sn^{4+} defects at the buried interface of Sn–Pb perovskite films, even under thermal stress. Consequently, the synergistic buried interface regulation of Co-SAM leads to a power conversion efficiency (PCE) of 23.46%, which outperforms devices modified with 2PACz or glycine alone. The Co-SAM-modified Sn–Pb PSC demonstrates enhanced thermal stability, maintaining 88% of its initial PCE under 65 °C thermal stress for 590 h.

KEYWORDS: co-self-assembled monolayers, hole selective layers, interfacial passivation, tin–lead perovskite solar cells, inverted perovskite solar cells



1. INTRODUCTION

Recent developments in organic–inorganic metal halide perovskite solar cells (PSCs) have yielded impressive results, with state-of-the-art single-junction PSCs achieving power conversion efficiencies (PCEs) exceeding 26%.¹ This progress is mainly attributed to the utilization of a pure lead (Pb) perovskite light absorber with a bandgap of 1.5–1.6 eV.^{2–4} However, in accordance with the Shockley–Queisser limit (S–Q limit), tin–lead (Sn–Pb) perovskites (~ 1.25 eV) have risen to prominence as leading contenders.^{5,6} Furthermore, these Sn–Pb perovskites are suitable absorbers for double⁷ and multijunction⁸ tandem PVs, which can maximize the S–Q limit to over 44%.

The hole selective layer (HSL), deposited between the perovskite absorber and the indium tin oxide (ITO) anode, exerts a profound influence on the PCE of inverted (p–i–n)

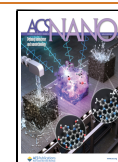
PSCs. Although poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate) (PEDOT:PSS) stands as the most commonly utilized hole selective material (HSM) in Sn–Pb PSCs, its undesirable acidic property poses challenges to achieving high efficiency and long-term stability in these devices.^{9–11} Additionally, another significant limiting factor is the nonradiative recombination originating from the bottom interface between the perovskite layer and the HSL.^{12,13} It is imperative to distinctly control the aforementioned buried

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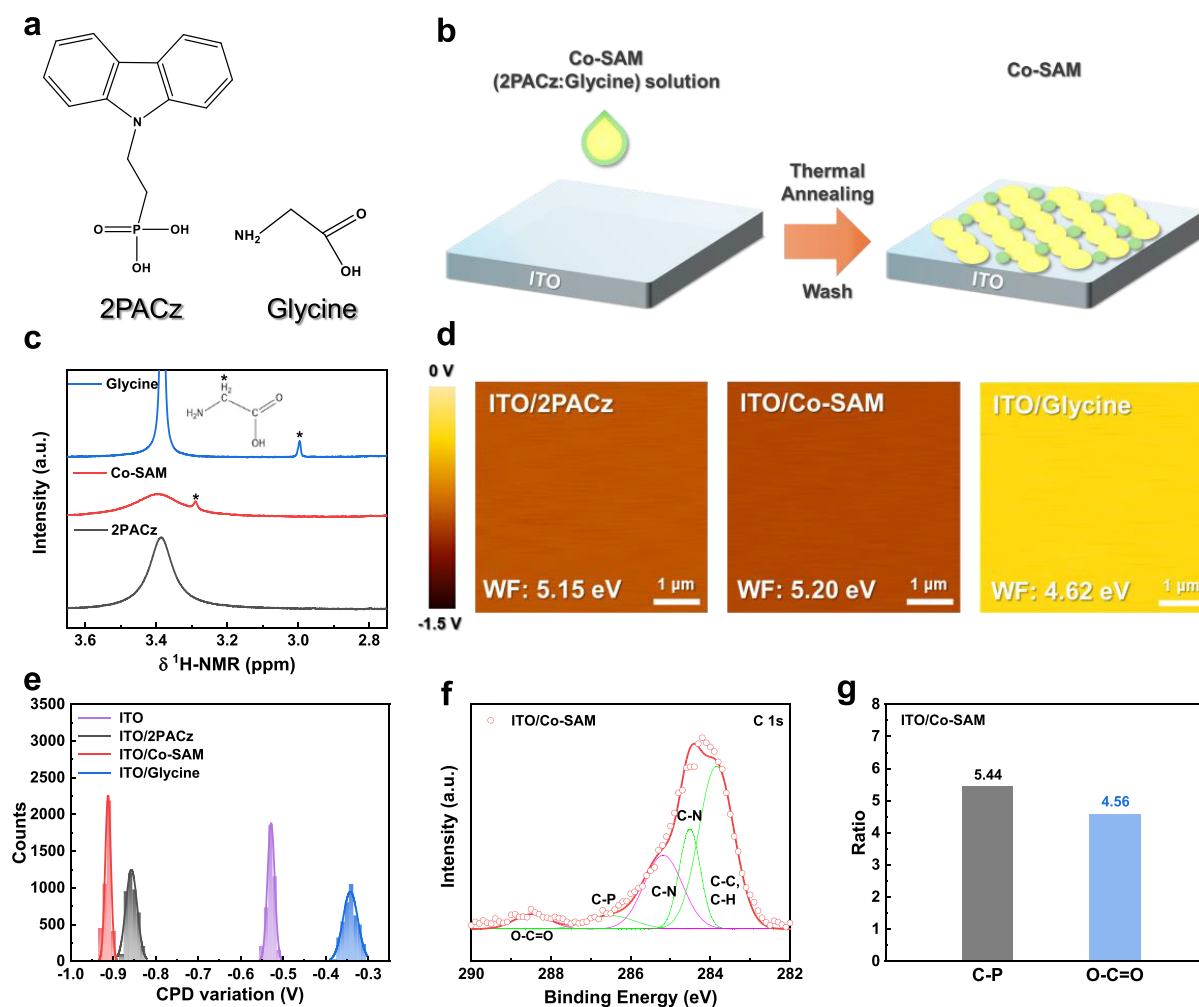


Figure 1. (a) Chemical structures of 2PACz and glycine. (b) Schematic diagram of one-step Co-SAM deposition. (c) ¹H NMR spectra of 2PACz, Co-SAM (2PACz:glycine), and glycine solution dissolved in DMSO-*d*₆. (d) Surface potential images of ITO/2PACz, ITO/Co-SAM, and ITO/glycine, and (e) the corresponding statistical contact potential difference variations obtained by KPFM. (f) C 1s XPS spectra of Co-SAM deposited on ITO and (g) the corresponding ratio of C–P and O–C=O bonds extracted from XPS spectra.

interface of Sn–Pb PSCs, where numerous Sn vacancies can form due to nonuniform nucleation and rapid crystallization.^{14,15} This leads to defective grain surfaces that are susceptible to trap formation and further Sn²⁺ oxidation.^{16,17}

Glycine hydrochloride (Gly-HCl) has been exclusively deployed as a bottom surfactant, targeting the passivation of defects present at the buried interface of the Sn–Pb PSCs. The incorporation of Gly-HCl, as an additive in the Sn–Pb perovskite precursor solution^{18,19} or PEDOT:PSS,²⁰ enhances hole extraction by forming a bottom surface dipole and passivating uncoordinated Sn in the perovskite film through its chloride ion (Cl[−]) and carboxylic group (−COOH). However, the introduction of Gly-HCl does not have the effect of stabilizing the interface between acidic PEDOT:PSS and Sn–Pb perovskites.

Recently, self-assembled monolayers (SAMs) have been effectively used as HSLs to replace PEDOT:PSS, benefiting from their easily tunable energy level alignment with perovskite and minimized parasitic absorption.²¹ The constituents of SAM molecules usually involve a functional head group and an anchoring group, such as phosphonic acid²² and carboxylic acid,^{23–25} which exhibit affinity toward binding with ITO substrates. The attainment of a highly dense SAM is pivotal for

ensuring a low-loss interface, optimizing light management and enhancing interface passivation concurrently.²⁶ In an effort to develop a suitable SAM for Sn–Pb PSCs, several different structures of SAM molecules have been proposed^{24,27} and an application of coating in a bilayer form.²⁸ Nevertheless, the various approaches proposed to enhance the performance of Sn–Pb PSCs have largely focused on the employment of reducing agents^{7,29–33} and posttreatment.^{18,34,35} Consequently, less attention has been paid to addressing the simultaneous construction of a conformal SAM and the passivation of the buried interface in Sn–Pb perovskites.

Herein, we propose a Co-Self-Assembled monolayer (Co-SAM) that can be applied by one-step deposition of a mixed processing solution containing both 2PACz and glycine. Glycine molecules are applied as SAM in Sn–Pb PSCs, modifying the buried interface through amine (−NH₂) and ammonium (−NH₃⁺) treatments, which impact on the bottom surface of Sn–Pb perovskites is yet to be unveiled. The interaction between 2PACz and glycine, as validated by ¹H nuclear magnetic resonance (NMR) spectroscopy, leading to the formation of a conformal SAM with uniform surface potential, characterized by Kelvin probe force microscopy (KPFM).^{21,26} Moreover, the −NH₂ and −NH₃⁺ groups in

glycine facilitate the vertical growth of the subsequent Sn–Pb perovskites and intimately interact with uncoordinated Sn to suppress the formation of Sn^{4+} defects at the interface even under continuous thermal stress. Through this synergistic buried interface regulation of Co-SAM, we achieved the second highest PCE of 23.46% in MA-free Sn–Pb PSCs, with high open-circuit voltage (V_{OC}) and fill factor (FF) values of 0.884 V and 80.76%, respectively. In addition, the Co-SAM-modified devices exhibited improved thermal stability, retaining 88% of the initial PCE after continuous exposure to 65 °C for 590 h.

2. RESULTS AND DISCUSSION

The molecular configurations of 2PACz and glycine are illustrated in Figure 1a. In 2PACz, there is a carbazole head group alongside a phosphonic acid anchoring group.²¹ Conversely, glycine features an amine head group and a carboxylic acid anchoring group,^{22,36} enabling to form SAM with an oriented dipole on ITO surface by a simple aqueous solution spin-coating method. The reaction of carboxylic acid of glycine with hydroxyl group present on ITO surface has been observed by the signal peaks of In 3d and Sn 3d shift toward a lower binding energy compared to the pristine ITO (Figure S1, Supporting Information).³⁷ In the case of the sole 2PACz-processing solution, the hydrophobic carbazole-based groups were oriented inwardly, while the hydrophilic phosphonic acids were exposed outwardly to the polar solvent environment.²¹ The resultant micelles were detected in the processing solutions through dynamic light scattering (DLS) (Figure S2, Supporting Information). Similarly, glycine molecules, when dissolved solely in *N,N*-dimethylformamide (DMF), exhibited clustering behavior facilitated by hydrogen bonding, resulting in the formation of dimers.³⁸ The agglomeration in their solution states ultimately led to the formation of an irregular and less dense SAM on the ITO substrate using the spin-coating method.^{21,26,39} To address these challenges, we employed a one-step deposition process involving 2PACz and glycine (referred to as Co-SAM) on ITO substrates through solution processing. This was followed by film annealing at 100 °C and solvent washing (Figure 1b).

Initially, to investigate the interaction between $-\text{NH}_2$ in glycine and phosphonic acid ($-\text{PO}(\text{OH})_2$) of 2PACz, we conducted ^1H NMR spectroscopy measurements. The ^1H NMR spectra of 1 mM solutions of 2PACz, Co-SAM (with a molar ratio of 2PACz to glycine = 5:5), and glycine are displayed in Figure 1c. The peak at $\delta = 3.00$ ppm, assigned to the hydrogen within $-\text{CH}_2$ of glycine, experienced an upfield shift to $\delta = 3.29$ ppm in the Co-SAM solution. This shift is linked to protonation of adjacent $-\text{NH}_2$. In the presence of both 2PACz and glycine, an acid–base reaction may occur between $-\text{NH}_2$ and $-\text{PO}(\text{OH})_2$, owing to their lower values of the negative logarithm of the base dissociation constant ($\text{p}K_{\text{b}}$) for alkyl-amines (~ 4.4)⁴⁰ and the negative logarithm of the acid dissociation constant for the first proton ($\text{p}K_{\text{a1}}$) of the phosphonic acid (~ 2.5).^{41,42} This result is further supported by the precise correlation observed between the degree of upfield shift and the ratios of 2PACz:glycine (1:9 and 7:3) in the mixed solutions (Figure S3, Supporting Information).

This interaction was further substantiated by Fourier transform infrared spectroscopy (FTIR) measurements. When glycine is dissolved, $-\text{NH}_2$ and $-\text{NH}_3^+$ groups coexist at the N-terminus and COO^- and COOH coexist at the C-terminus in the solution state depending on the pH of the

solution.^{43,44} In Figure S4a (Supporting Information), an absorption peak appears at 1725 cm^{-1} was shown, suggesting the presence of both C- and N-termini protonated glycine only in the Co-SAM solution. In the Co-SAM solution, when 2PACz and glycine are mixed, the equilibrium shifts to a more protonated glycine, resulting in a decrease of COO^- (at $\sim 1600\text{ cm}^{-1}$) and an increase of $\text{H}_3\text{N}^+-\text{CH}_2-\text{COOH}$ (at $\sim 1725\text{ cm}^{-1}$). The difference signal shown in Figure S4b corroborates this. In the Co-SAM solution, the proton source that can generate $\text{H}_3\text{N}^+-\text{CH}_2-\text{COOH}$ is the phosphonic acid ($\text{PO}(\text{OH})_2$) group of 2PACz, so a significant amount of $-\text{NH}_3^+$ and $-\text{PO}_2(\text{OH})^-$ coexist in the solution. Consequently, the electrostatic interaction between $-\text{NH}_3^+$ of glycine and $-\text{PO}_2(\text{OH})^-$ of 2PACz is expected, which may result in the inhibition of micelle formation of both 2PACz and glycine (Figure S5, Supporting Information). As a result of this, Co-SAM readily forms a uniform monolayer because glycine can attach to the vacancies within the 2PACz structure, ensuring more complete coverage.⁴⁵ Hence, when analyzing atomic force microscopy (AFM) images, it was observed that the surface roughness of ITO/Co-SAM (4.25 nm) was smoother than the ITO/2PACz (4.66 nm) or ITO/glycine (4.31 nm) as well as the bare ITO (4.29 nm) (Figure S6, Supporting Information). This implies that Co-SAM leads to a smoother ITO surface and contributes to diminishing surface defects.

KPFM was utilized to assess the surface contact potential differences (CPDs) and the corresponding statistical variations of ITO/2PACz, ITO/Co-SAM, and ITO/glycine (Figure 1d). The work function (WF) values of ITO substrates modified by 2PACz, Co-SAM, and glycine were measured at 5.15, 5.20, and 4.62 eV, respectively. Notably, the WF of ITO/Co-SAM closely followed the trend of 2PACz. This alignment can be attributed to the carboxylic group in glycine, which exhibits a lower anchoring ability compared to phosphonic acid due to its weaker acidity.⁴⁶ Meanwhile, the WF of ITO/Co-SAM was deeper than that of ITO/2PACz, indicating improved packing quality of Co-SAM, which may contribute to an improvement in V_{OC} .^{21,47} Moreover, ITO/Co-SAM showed improved uniformity in the surface potential, as indicated by narrower CPD variation (Figures 1e).

The optimal ratio of 2PACz:glycine in the Co-SAM solution is 5:5, which has been determined based on the device performances (Figure S7, Supporting Information). The composition of the Co-SAM on the ITO was determined by C 1s characterization of X-ray photoelectron spectroscopy (XPS) spectra, as shown in Figures 1f,g and S8 (Supporting Information). The peaks of C–P (2PACz) and O–C=O (glycine), located at binding energy of 286.93²² and 288.48 eV⁴⁸, respectively, were detected simultaneously. Integrating the peak area of the C 1s peaks⁴⁶ yielded a ratio of 2PACz to glycine of approximately 5.44:4.56, which slightly differs from that of its solution state (Table S1, Supporting Information), due to the acidity difference between anchoring groups. Furthermore, the presence of Co-SAM molecules anchoring on the ITO substrate without diffusion into adjacent interlayers during crystallization or operation is fully validated by time-of-flight secondary ion mass spectrometry (ToF-SIMS) depth profiles (Figure S9, Supporting Information).

To examine the impact of incorporating Co-SAM as HSLs after Sn–Pb perovskite film coating, scanning electron microscopy (SEM) was employed to observe the morphology of the bottom surface of the delaminated perovskite films from SAM-modified ITO substrates.^{13,45} On the bottom surface of

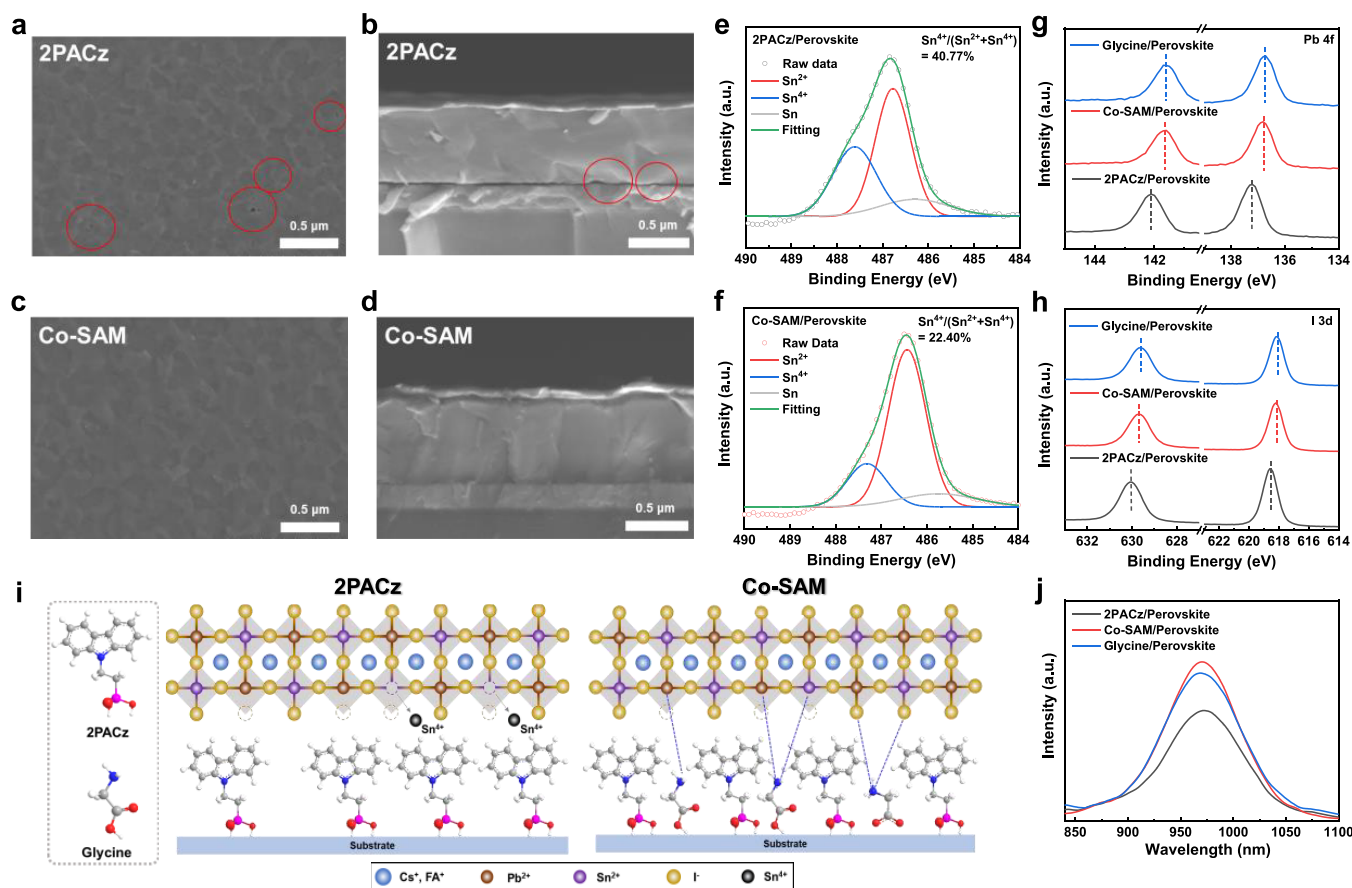


Figure 2. Buried interface and cross-sectional SEM images of perovskite films coated on (a,b) ITO/2PACz and (c,d) ITO/Co-SAM. Sn 3d_{5/2} XPS spectra of perovskite films coated on (e) ITO/2PACz and (f) ITO/Co-SAM after aging at 65 °C for 40 h. (g) Pb 4f and (h) I 3d XPS spectra of perovskite films coated on ITO/2PACz, ITO/Co-SAM, and ITO/glycine. (i) Schematic illustration of the lattice structure of Sn–Pb perovskite films coated on ITO/2PACz and ITO/Co-SAM. (j) Steady-state PL spectra of perovskite films deposited on ITO/2PACz, ITO/Co-SAM, and ITO/glycine.

perovskite films deposited on ITO/2PACz, several small pore defects at the grain boundary were observed, as indicated by the red solid circles (Figure 2a,b). The presence of such voids at the bottom of perovskite films could potentially impede hole extraction at the interface, consequently exacerbating non-radiative recombination.⁴⁹ In contrast, the bottom surfaces of the films coated on ITO/Co-SAM and ITO/glycine showed pinhole-free and uniform morphology (Figures 2c and S10, Supporting Information), which is also proven by vertically well-arranged grains, as shown in cross-sectional SEM images (Figures 2d and S11, Supporting Information).

To investigate the crystallinity of the perovskite according to the SAM-modified substrate, X-ray diffraction (XRD) analysis was conducted. It showed that the perovskite grown on ITO/Co-SAM exhibited stronger (100) and (200) diffraction peaks than that grown on ITO/2PACz (Figure S12, Supporting Information).²⁷ By fitting the (100) diffraction peaks, the full width at half-maximum (FWHM) of the peaks was determined to be 0.11 and 0.10 for 2PACz and Co-SAM-based perovskite, respectively. Furthermore, grazing incidence wide-angle X-ray scattering (GIWAXS) images demonstrates two prominent scattering peaks, which are centered on $q_z \approx 1.0 \text{ \AA}^{-1}$ and $q_z \approx 2.0 \text{ \AA}^{-1}$, corresponding to the (100) and (200) crystal planes⁵⁰ (Figure S13, Supporting Information), as evidenced in the XRD result. Compared to the perovskite coated on ITO/2PACz, the more intense and narrower scattering peaks for

(100) and (200) planes are shown in that deposited on the ITO/Co-SAM. This result may be attributed to the templating effect of $-\text{NH}_2/-\text{NH}_3^+$, which may potentially anchor I/I⁻ inside perovskite precursor PbI_2 and SnI_2 through hydrogen bonding, thereby facilitating the growth of crystal.⁵¹

The oxidation of Sn^{2+} is accelerated by iodide vacancies (V_I) and the oxidation of I^- to I_2 , with numerous defects such as V_I and tin or lead vacancies (V_{Sn} or V_{Pb}) concentrated at the interfaces of the perovskite layer.^{52,53} In case of Sn–Pb PSCs, defective buried surfaces with numerous V_I and V_{Sn} , which are susceptible to further Sn^{2+} oxidation, are prone to be formed due to nonuniform nucleation and rapid crystallization.^{14,16} The occurrence of such oxidation at the buried interface was simulated using XPS on Sn–Pb perovskite films with low solution concentration (0.1 M) deposited on ITO/SAM.⁵⁴ The Sn 3d_{5/2} peak exhibited two distinct peaks at 486.5 and 487.4 eV,⁷ corresponding to Sn^{2+} and Sn^{4+} , respectively. The Sn 3d_{5/2} XPS spectra (Figure S14, Supporting Information) and the ratio of Sn^{4+} , 20.78% (Table S2, Supporting Information), demonstrated a substantial presence of Sn^{4+} at the perovskite film deposited on ITO/2PACz. In contrast, Sn^{4+} defects were effectively suppressed at that deposited on ITO/Co-SAM (17.55%), a slight reduction compared to the ratio observed when ITO/glycine was employed (17.79%).

In operational conditions, thermal stress is inevitable for Sn–Pb PSCs. However, it decreases the activation energy

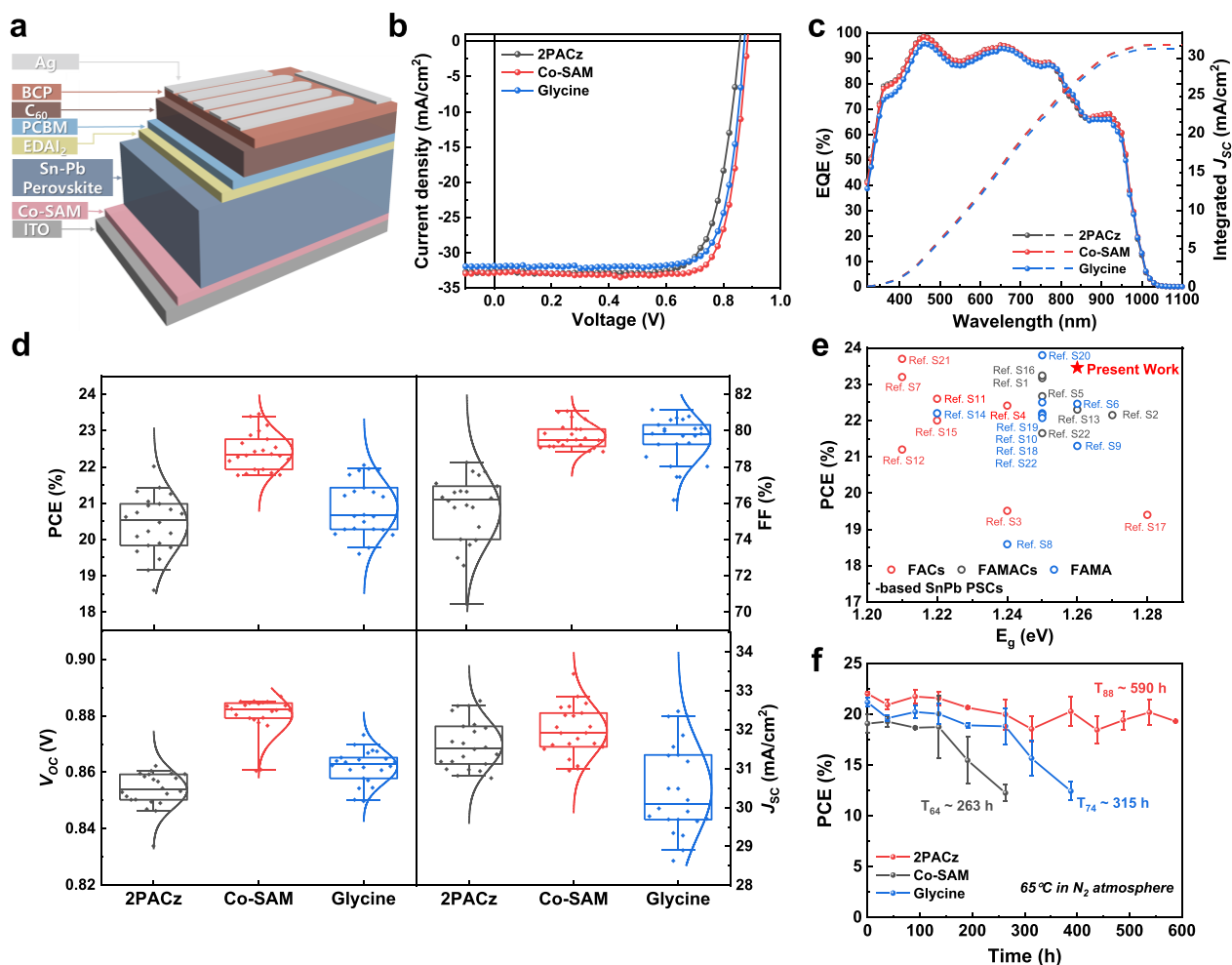


Figure 3. (a) Device structure of the Sn–Pb PSC. (b) J – V curves of PSCs modified by 2PACz, Co-SAM, and glycine, with derived J_{SC} , V_{OC} , FF, and PCE. (c) EQE spectra of the best PSCs. (d) Statistical distribution of PV parameters of PSCs. (e) Plot showing the PCE and bandgap (E_g) values of previously reported Sn–Pb PSCs fabricated on ITO; detailed parameters are listed in Table S5 (Supporting Information). (f) Thermal stability of unencapsulated Sn–Pb PSCs modified by 2PACz, Co-SAM, and glycine at 65 °C under dark conditions in a N_2 environment.

required for Sn^{4+} defect formation, even under inert conditions, with a temperature of 65 °C corresponding to $kT \approx 29$ meV. The Sn^{4+} defects could form SnO_2 on all interfaces, which might significantly impede charge carrier extraction, particularly in cases where the interface between the Sn–Pb perovskite and HSL is compromised.⁵³ Figure 2e,f illustrates the Sn 3d_{5/2} XPS spectra and fitting curves for the bottom surface of Sn–Pb perovskite film coated on ITO/2PACz and ITO/Co-SAM after aging at 65 °C for 40 h. The effective suppression of surface-activated corrosion was corroborated by the reduction in the Sn^{4+} ratio of the perovskite film deposited on ITO/Co-SAM (22.40%), compared to that deposited on ITO/2PACz (40.77%) (Table S3, Supporting Information), which was even more pronounced than that deposited on ITO/glycine (25.86%) (Figure S15, Supporting Information).

We postulate that suppression of Sn^{4+} defect formation at the buried surface of Sn–Pb perovskite films is due to an interaction between glycine and perovskite. To investigate further, we conducted additional Pb 4f and I 3d XPS spectra on perovskite films deposited on ITO/SAM (Figures 2g,h and S16, Supporting Information). In the case of perovskite films deposited on ITO/Co-SAM and ITO/glycine, the binding energies of both the metal cations and the halide ions are

shifted²⁰ compared to those of the perovskite film deposited on ITO/2PACz. It was demonstrated that the electron-donating $-NH_2$ ³⁵ in glycine could passivate the uncoordinated Pb and Sn.^{24,55,56} In addition, $-NH_3^+$ actively inhibited the generation of V_I through hydrogen bonding ($N-H \cdots I$)^{57,58} and further mitigated the defect formation at the buried interface of Sn–Pb PSCs, even under continuous thermal stress. Figure 2i illustrates the proposed mechanisms of synergistic buried interface regulation via Co-SAM compared to Sn–Pb perovskite films coated on ITO modified by 2PACz only.

Ultraviolet photoelectron spectroscopy (UPS) was employed to ascertain the energy level alignment between the Sn–Pb perovskite layer and ITO/SAM. The resulting energy level diagrams of ITO/SAM/Sn–Pb perovskite bilayers are depicted in Figure S17 (Supporting Information). A vacuum level shift of 0.26 eV was observed for ITO/2PACz/perovskite, which further increased to 0.39 eV for ITO/Co-SAM/perovskite. This resulted in enhanced carrier extraction, attributed to the enlarged built-in field.^{59,60} However, the extraction of holes was anticipated to be challenging in the case of ITO/glycine/perovskite due to a significant energy barrier. Consequently, the Sn–Pb perovskite film deposited on ITO/

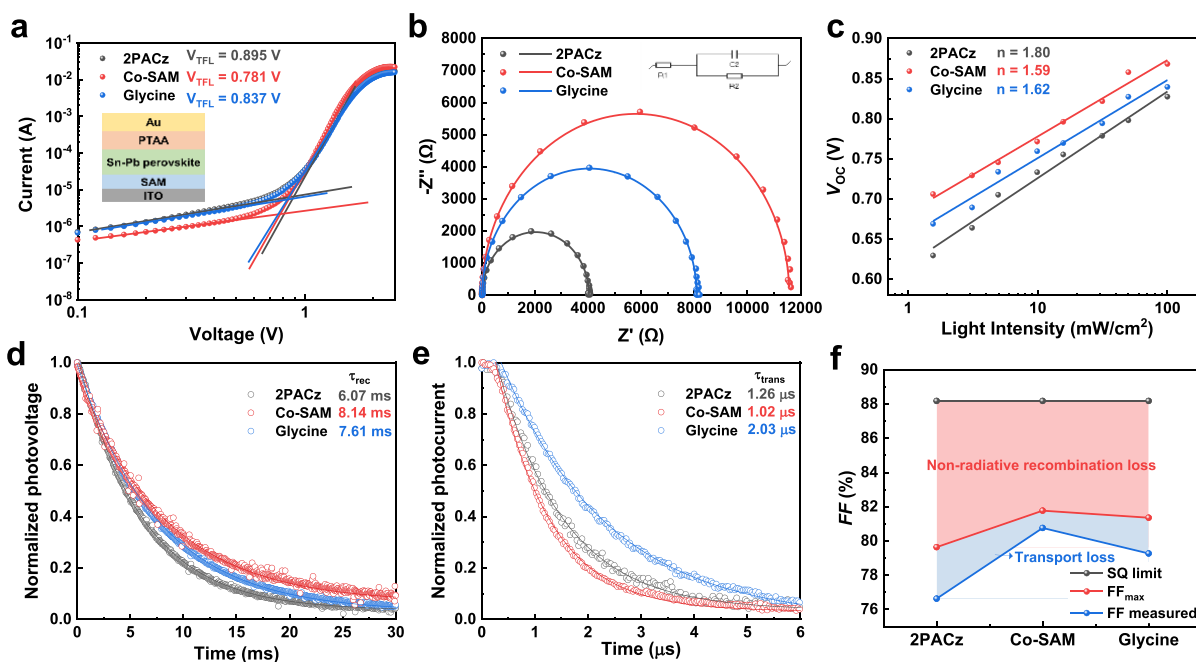


Figure 4. (a) SCLC characteristics of hole-only devices with 2PACz, Co-SAM, and glycine. (b) Nyquist plot for Sn–Pb PSCs with 2PACz, Co-SAM, and glycine. (c) Light dependence of V_{oc} of Sn–Pb PSCs with 2PACz, Co-SAM, and glycine. (d) Transient photovoltage measurements and (e) transient photocurrent measurements of Sn–Pb PSCs with 2PACz, Co-SAM, and glycine. (f) FF loss analysis of devices with 2PACz, Co-SAM, and glycine.

Co-SAM exhibited a higher photoluminescence (PL) intensity compared to those of ITO/2PACz and ITO/glycine (Figure 2j), indicating a reduction in nonradiative recombination at the buried interface. This is attributed to the passivation of defects and the minimization of the energy barrier for hole extraction.^{46,61,62}

The film morphologies were characterized by AFM and SEM, as shown in Figures S18 and S19 (Supporting Information). The perovskite grown on ITO/Co-SAM exhibited lower roughness (22.35 nm) compared to those on ITO/2PACz (23.13 nm) and ITO/glycine (22.60 nm), indicating a more homogeneous crystallization on the surface.⁴⁶ The top view SEM images of perovskite films deposited on an ITO/SAM exhibited no apparent changes. To analyze the influence of Co-SAM on the optoelectronic properties of the Sn–Pb perovskite films, we conducted UV–vis spectra. The absorbance spectra and the Tauc plot in Figure S20 (Supporting Information) demonstrated negligible differences in absorbance and band gap (1.26 eV) between the perovskite films coated on different SAM.

Moreover, CPDs and the corresponding variations of perovskite films coated on different SAM were confirmed by KPFM. The deposition of Sn–Pb perovskite on ITO/Co-SAM resulted in a noticeable reduction in the fluctuations observed in the CPD (Figure S21a,b, Supporting Information).^{63,64} The decreased WF of the Co-SAM-based perovskite can facilitate charge transport at the interface between perovskite and C_{60} (Figure S21c, Supporting Information).^{65,66}

The synergistic buried interface regulation through Co-SAM ensures an energetically well-regulated interface, facilitating hole extraction and substantially reducing interfacial defect states. To analyze the influence of 2PACz, Co-SAM, and glycine on the photovoltaic performance of devices, we fabricated Sn–Pb PSCs based on these SAMs with the configuration of ITO/SAM/ $\text{FA}_{0.8}\text{Cs}_{0.2}\text{Sn}_{0.5}\text{Pb}_{0.5}\text{I}_3$ /ethane-1,2-

diammonium iodide (EDAI_2)/phenyl-C61-butyric-acid-methyl-ester (PCBM)/ C_{60} /bathocuproine (BCP)/Ag (Figure 3a). The current density–voltage (J – V) curves for the champion devices with 2PACz, Co-SAM, and glycine are shown in Figure 3b, with the corresponding photovoltaics parameters summarized in Table S4 (Supporting Information). The champion device based on Co-SAM exhibited a PCE of 23.46%, with a V_{oc} of 0.884 V, a J_{sc} of 32.85 mA cm^{-2} , and a FF of 80.76%, which demonstrates the combined effects of 2PACz and glycine. The 2PACz-based champion device displays a PCE of 21.43% with a V_{oc} of 0.857 V, a J_{sc} of 32.61 mA cm^{-2} , and an FF of 76.63%. The glycine-based champion device exhibits a PCE of 22.05% with a V_{oc} of 0.873 V, a J_{sc} of 31.86 mA cm^{-2} , and a FF of 79.27%. In the external quantum efficiency (EQE) spectra (Figure 3c), the integrated J_{sc} values are 31.37 (2PACz), 31.43 (Co-SAM), and 30.91 mA cm^{-2} (glycine), which aligns well with the J_{sc} values obtained from J – V characterization. The corresponding distributions of solar cell parameters are summarized in Figure 3d, which demonstrates the excellent reproducibility for the devices modified by Co-SAM. The synergistic effects of Co-SAM contribute to a reduction in buried interface defects, leading to elevated V_{oc} and FF. The Sn–Pb PSC with Co-SAM which derived the best $V_{oc} \times \text{FF}$ displays $0.903 \text{ V} \times 80.54\%$ (Figure S22, Supporting Information). As illustrated in Figure 3e, the optimal photovoltaic performance of Co-SAM-based Sn–Pb PSCs represents the second highest value among previously reported MA-free Sn–Pb PSCs fabricated on ITO (Table S5, Supporting Information).

The hysteresis index (HI) of Co-SAM-treated devices is reduced by 9.14% in comparison to those of 2PACz-treated devices (10.09%) and glycine-treated devices (9.37%) (Figure S23 and Table S6, Supporting Information). The stabilized output PCEs under the maximum power point are 21.9% at 0.746 V, 21.0% at 0.723 V, and 17.3% at 0.650 V for the Co-

SAM, glycine, and 2PACz-based devices for 100 s (Figure S24, Supporting Information). The devices modified with Co-SAM demonstrated considerably enhanced stability, with 91.8% of its maximum PCE remaining after 1680 h of storage in a nitrogen atmosphere (Figure S25, Supporting Information). In comparison, the devices modified with 2PACz and glycine maintained 87.6 and 91.6% of their highest PCE, respectively. The observed reduction in PCE can be attributed to a reduction in FF, while the commendable stability can be attributed to the advantageous interface contact established between Co-SAM and Sn–Pb perovskite, effectively retarding the degradation rate.^{67,68}

Moreover, under continuous thermal exposure for 590 h at 65 °C, the Sn–Pb PSC modified by Co-SAM maintained 88% of its initial PCE, exhibiting enhanced thermal stability than those modified by 2PACz or glycine only (Figures 3f and S26, Supporting Information). The devices based on 2PACz and glycine retained 64 and 74% of their initial PCEs after 263 and 315 h, respectively. It can be observed that interfacial passivation by Co-SAM is more effective in restraining Sn⁴⁺ defects formation when the devices are exposed to vulnerable environments. The Sn–Pb perovskite films modified by 2PACz, which lack –NH₂ or –NH₃⁺ passivation at the buried interface, are susceptible to defect formation following prolonged heat exposure, as evidenced by the XPS data. On the other hand, the thermal stability of glycine-modified devices is constrained by comparatively weaker anchoring capacity of the carboxylic group.⁴⁶ This phenomenon can be attributed to the potential for the bond between the anchoring group of SAM-forming molecules and substrates to be broken through desorption by thermal stress.⁶⁹

To understand the improved photovoltaic performance of devices incorporating Co-SAM compared to those utilizing solely 2PACz or glycine, space charge limited current (SCLC) measurements were conducted on devices with a hole-only structure (ITO/SAM/Sn–Pb perovskite/poly(triaryl amine) (PTAA)/Au) (Figure 4a). The measured trap filled limit voltage (V_{TFL}) values are 0.895 for 2PACz, 0.781 for Co-SAM, and 0.837 V for glycine-based devices, corresponding to trap densities of 8.94×10^{15} , 7.80×10^{15} , and $8.83 \times 10^{15} \text{ cm}^{-3}$, respectively.⁷⁰ The minimal density of traps observed in the device modified by Co-SAM indicates a notable reduction in defects at Co-SAM/Sn–Pb perovskite interface.⁷¹

To investigate the effect of Co-SAM on the charge dynamics, electrochemical impedance spectroscopy (EIS) analysis was performed. The resulting Nyquist plot and the corresponding equivalent circuit are shown in Figure 4b. The Co-SAM-modified device exhibits a higher recombination resistance (R_{rec}) (11467 Ω) than that of the device with 2PACz (4023 Ω) due to the suppression of interfacial recombination passivated by glycine.⁷² As shown in Figure 4c, we also obtained the ideal factor (n) for the 2PACz, Co-SAM, and glycine devices by fitting the V_{OC} dependence on light intensity, which is shown to be 1.80, 1.59, and 1.62, respectively. The smaller n value of the Co-SAM-based device than that of the 2PACz-based device suggests that trap-assisted recombination is effectively suppressed, showing a trend similar to that of glycine-based devices.

To further understand the improvement in V_{OC} of devices with Co-SAM, Mott–Schottky analysis was conducted via capacitance–voltage measurements, as shown in Figure S27 (Supporting Information). The device with 2PACz exhibits a built-in potential (V_{bi}) of 0.576 V, while the devices modified

by Co-SAM and glycine show higher V_{bi} values of 0.633 and 0.615 V, respectively. This increase in V_{bi} suggests that the incorporation of glycine improves interfacial carrier transport, which leads to enhanced V_{OC} .⁷³

A log–log plot of J_{SC} as a function of the light intensity and its fitting curve suggest a larger α (slope of curve) value for the Co-SAM-modified device (0.992) than for the device with glycine (0.961), implying that charge carrier recombination is reduced and higher J_{SC} can be achieved⁷⁰ (Figure S28, Supporting Information). Furthermore, the improved charge transport behavior for the Co-SAM-modified Sn–Pb PSCs was observed through photovoltage (TPV) and transient photocurrent (TPC) measurements (Figure 4d,e). The prolonged photovoltage lifetime of Co-SAM-based devices (8.14 ms) compared to the 2PACz-based devices (6.07 ms) implies that a smaller number of carriers were trapped at the buried interface due to the presence of glycine. The rapid carrier extraction time of 2PACz-modified devices (1.26 μs) contributes to that of Co-SAM-modified devices (1.02 μs), rather than that of glycine-modified devices (2.03 μs).

We investigated the FF loss of Sn–Pb PSCs (Figure 4f). The maximum FF (FF_{max}), which corresponds to no loss in charge transport, was calculated using the following equation where n , k , T , and q represent the ideality factor, Boltzmann constant ($1.381 \times 10^{-23} \text{ m}^2 \text{ kg s}^{-2} \text{ K}^{-1}$), temperature (300 K), and elementary charge ($1.602 \times 10^{-19} \text{ C}$), respectively.⁷⁴

$$\text{FF}_{\text{max}} = \frac{v_{\text{OC}} - \ln(v_{\text{OC}} + 0.72)}{v_{\text{OC}} + 1} \quad (\text{where } v_{\text{OC}} = \frac{qV_{\text{OC}}}{nkT})$$

The calculated FF_{max} values of the Sn–Pb PSCs with 2PACz, Co-SAM, and glycine are 79.64, 81.77, and 81.36%, respectively. The increased FF_{max} aligns well with the decreased nonradiative recombination observed in the Co-SAM-treated device. In the Co-SAM-based devices, the FF loss induced by charge transport is minimal, at 1.01%. In contrast, the 2PACz and glycine-modified devices exhibit FF losses of 3.01 and 2.09%, respectively.

3. CONCLUSIONS

In conclusion, we have focused on improving the performance and stability of Sn–Pb PSCs through the introduction of Co-SAM by mixing 2PACz and glycine. This approach successfully minimized the formation of micelles due to the interaction between 2PACz and glycine, ensuring a conformal SAM on ITO. In addition, it proposed –NH₂ and –NH₃⁺ passivation at the bottom surface of Sn–Pb perovskite films by glycine, leading to the suppression of Sn⁴⁺ defects. The Co-SAM strategy improved the quality of the subsequently coated Sn–Pb perovskites, resulting in an improved morphology and crystallinity. Consequently, the Co-SAM-modified Sn–Pb PSCs achieved an impressive PCE of 23.46% with improved V_{OC} and FF due to the improved interfacial quality and reduced defects. In addition, the Co-SAM-modified devices exhibited outstanding thermal stability, retaining 88% of the initial PCE after prolonged exposure at 65 °C for 590 h. This research allows for further development of highly efficient and reliable Sn–Pb PSCs.

4. MATERIALS AND METHODS

4.1. Materials. All the chemicals and solvents used in preparing perovskite precursor solution were used as they were bought without any purification. Formamidinium iodide (FAI, 99.99%) and ethane-

1,2-diammonium iodide (EDAI₂) were purchased from Greatcell Solar Materials. *N,N*-Dimethylformamide (DMF, anhydrous 99.8%), dimethyl sulfoxide (DMSO, 99.9%), cesium iodide (CsI, 99.9%), tin(II) iodide (SnI₂, 99.99%), tin(II) fluoride (SnF₂, 99%), Sn powder (99.5%), lead thiocyanate (Pb(SCN)₂, 99.5%), glycine (>98.5%), 2-propanol (IPA, anhydrous, 99.5%), chlorobenzene (CB, 99.8%), and toluene (anhydrous, 99.8%) were purchased from Sigma-Aldrich. ([2-(9*H*-Carbazol-9-yl) ethyl] phosphonic acid (2PACz, >98%), lead iodide (PbI₂, 99.99%), and bathocuproine (BCP, 99%) were purchased from TCI. C₆₀ was purchased from Seecchem. Phenyl-C61-butyric-acid-methyl-ester (PCBM, 99.5%) was purchased from OSM. Indium tin oxide on glass has 10 Ω sq⁻¹.

4.2. Perovskite Precursor Solution Preparation.

FA_{0.8}CS_{0.2}Sn_{0.5}Pb_{0.5}I₃ perovskite precursor was prepared inside a N₂-filled glovebox. 247.64 mg of FAl, 88.86 mg of CsI, 335.27 mg of SnI₂, 414.91 mg of PbI₂, 14.1 mg of SnF₂, 2.91 mg of Pb(SCN)₂, and 5 mg of Sn powder were dissolved in 1 mL of the DMF/DMSO mixture (4:1 volume ratio). Perovskite precursor was stirred for overnight. The solution was filtered with a 0.45 μm PTFE filter before use.

4.3. Device Fabrication. ITO patterned glass substrates were sequentially cleaned in deionized water, acetone, and isopropyl alcohol by ultrasonication and then dried in an oven at 80 °C. The ITO substrates were subjected to UV-ozone treatment for 30 min. 100 μL of 2PACz, glycine, or Co-SAM solutions (1 mM in DMF) were dropped on the substrates and spin-coated at 4000 rpm for 30 s. The self-assembly monolayer (SAM)-coated films were transferred onto a hot plate annealing at 100 °C for 10 min after DMF washing. The perovskite films were spin-coated at 5000 rpm for 45 s. 200 μL of CB was dropped onto the substrates at 20 s before spin coating was finished. Then, the perovskite films were treated on the hot plate at 100 °C for 10 min. Posttreatment was carried out by spin-coating solution of 0.5 mg mL⁻¹ EDAI₂ in IPA:toluene (1:1 volume ratio) at 4000 rpm 20s, followed by annealing at 100 °C for 5 min. PCBM (3 mg/mL in CB) was spin-coated at 3000 rpm for 30 s, followed by annealing at 70 °C for 5 min. All processes were conducted in the N₂-filled glovebox. Finally, 20 nm of C₆₀, 5 nm of BCP, and 100 nm of Ag were sequentially deposited on top of the perovskite layer by thermal evaporation (10⁻⁷ mbar).

4.4. Film Characterizations. Photoelectron spectroscopy was performed in an ultrahigh vacuum system, with a base pressure <10⁻⁹ mbar. The UPS and XPS were conducted with an ESCALAB 250XI. XRD patterns were obtained using a normal X-ray diffractometer (D8 ADVANCE, Bruker AXS) with the following settings: an angular range of 10–130° (2θ), a copper X-ray source, an X-ray generation energy of 3 kW (40 kV potential, 40 mA current), using a θ–θ based goniometer, and a graphite monochromator. Top view and cross-section structures of perovskite films were characterized by SEM (SU7000, Hitachi High-Technologies) at an accelerating voltage of 7 kV. The surface morphology of the films was characterized by AFM using a Dimension ICON (Bruker Nano Surface). Samples were prepared under optimized conditions, and the measurement was carried out in scanAsyst mode. The AFM was operated in tapping mode with scan range of 5 μm × 5 μm. The CPD between the ITO and Perovskite films before and after SAM treatment was measured by using kelvin probe force microscopy (KPFM) (SII Nano Technology Inc.). The topography and potential were simultaneously measured using two independent lock-in amplifiers, utilizing the cantilever resonance frequency as the feedback. The probe used was a Rh-coated Si cantilever with a resonance frequency of 25 kHz. DLS measurements were obtained from a Zetasizer Nano S/ZS instrument (Malvern Instruments Limited, U.K.) equipped with a 4.0 mW He–Ne laser at a wavelength of 633 nm. The scattered light was measured at an angle of 173° using a noninvasive backscatter method. Measurements were conducted at 25 °C using glass cuvettes. Solutions were prepared with DMF as the solvent at a concentration of 1 mM. To ensure reproducibility, measurements were repeated three times. Steady-state photoluminescence (PL) spectra signals were recorded using a high-sensitivity spectrophotometer (MAYA 2000 Pro, Ocean Insight Inc., Orlando, FL, USA). The wavelength of the PL excitation source was 632 nm. The FTIR spectra were

recorded with a resolution of 1 cm⁻¹ using an IRtracer-100 (Shimadzu Co.) equipped with an MCT detector under an N₂ gas purge environment. The samples' solvent was DMSO instead of DMF, which allowed us to observe peaks in the spectral range of 1500–1800 cm⁻¹. The IR cell comprised two CaF₂ windows (25 mm diameter and 2 mm thick) separated by 100 μm with a Teflon spacer. For the solution sample of 2PACz in DMSO, the concentration was ~5 mM. Due to solubility, saturated supernatants were used for solution samples of Co-SAM and glycine after centrifugation. To normalize the concentrations of the samples, each of the two spectra (Co-SAM and glycine) was multiplied by a scaling factor. Each of the four specimens (three solution samples and one DMSO solvent) underwent 512 scans, consisting of four sets of 128 scans. All FTIR spectra were recorded at 22 °C. The ¹H NMR spectroscopy was carried out using a 400 MHz FT-NMR spectrometer (AVANCE III HD, Bruker). GIWAXS measurements were conducted at the PLS-II U-SAXS beamline 6D of Pohang Accelerator Laboratory. X-ray light coming from the in-vacuum undulator (IVU) was monochromated ($E_k = 11.564$ keV, $\lambda = 1.07212$ Å) using a Si (111) double crystal monochromator and focused horizontally and vertically at the sample position (450 (*H*) × 60 (*V*) μm² in full width at half-maximum) using a K–B-type focusing mirror system. The incidence angle (0.50°) of the X-ray beam was chosen to allow for complete penetration into the film. GIWAXS patterns were recorded by using a 2D CCD detector (SX165, Rayonix, USA). The X-ray irradiation time was 20–60 s depending on the saturation level of the detector. Samples for GIWAXS measurements were prepared on ITO glass substrates.

4.5. Device Characterization. Current density–voltage (*J*–*V*) characteristics of solar cells were measured using a Keithley 2635A source measurement units inside the N₂-filled glovebox using a high-quality optical fiber to guide the light from a solar simulator to the device under AM 1.5G illumination at 100 mW cm⁻². *J*–*V* characteristics were measured with a scanning speed of 1 V s⁻¹. Stabilized power output was measured by continuously tracking the current density with the voltage at the MPP under AM 1.5G illumination. The EQE measurements were carried out for the solar cells using a QEX7 system (PV Measurement, Inc.). For *J*–*V* and EQE measurements, MgF₂ antireflection coating was applied to the PSCs. EIS measurements were conducted with a potentiostat (VMP3 and VSP-300, BioLogic) over range from 0.1 Hz to 1 MHz under dark conditions. The Mott–Schottky curves were measured using a potentiostat (VMP3/VSP-300, BioLogic) in the 0–1.1 V voltage range and at a frequency of 0.5 MHz under dark conditions. Transient photocurrent (TPC) and transient photovoltage (TPV) measurements were conducted using the analyzer function of an organic semiconductor parameter test system (McScience T4000) at *V*_{OC} conditions under 1sun illumination.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsnano.4c06396>.

XPS data, DLS particle size distribution, ¹H NMR spectra, FTIR spectra, molecular interaction images, AFM images, statistical device performance, ToF-SIMS depth profiles, SEM images, XRD patterns, GIWAXS images, UPS data, absorption spectra, KPFM images and statistical variations of CPD, *J*–*V* curves, hysteresis index, SPO of devices, long-term shelf life stability at N₂, thermal stability, Mott–Schottky plot, light dependence of *J*_{SC}, table for composition of Co-SAM, tables for calculation of Sn⁴⁺ ratio, and table for device performance (PDF)

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Author Contributions

J.R. and J.G.S. contributed equally to this work. J.R. performed the fabrication and J.R. and J.G.S. performed analysis of the devices and wrote the manuscript. S.P. and S.C. carried out KPFM, TPC, and TPV measurement. J.S. carried out ^1H NMR measurement. T.S. carried out SEM measurement. J.K. carried out DLS measurement. S.O.O. carried out XRD measurement. Y.J. carried out UV–vis absorption measurement. Y.L. carried out AFM measurement. Y.S.S. carried out steady-state PL measurement. H.J. and D.L. carried out EIS measurement. D.Y. carried out GIWAXS measurement. J.G.S. and Y.S.K. analyzed FTIR measurement. D.S.K. and J.Y.K. conceived and supervised the project.

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Notes

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ABBREVIATIONS

CCR2, CC chemokine receptor 2; CCL2, CC chemokine ligand 2; CCR5, CC chemokine receptor 5; TLC, thin layer chromatography.

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